

AVCDL Traceability

Revision

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Abstract

This document provides a visual overview of how the **AVCDL** achieves the traceability needed when creating a safety-critical, cyber-physical system.

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Overview

The creation of safety-critical, cyber-physical systems is a highly complex endeavor. An important aspect of such systems is the ability to ensure that every activity is supported by those preceding it in a manner which allows for tracing each decision back to the stated goals of the system. The AVCDL was constructed with this need in mind.

AVCDL Product Dependencies

The following diagram shows the end-to-end AVCDL phase product dependencies. These are presented as a dependency graph (reversed) using ISO standard flowchart symbols.

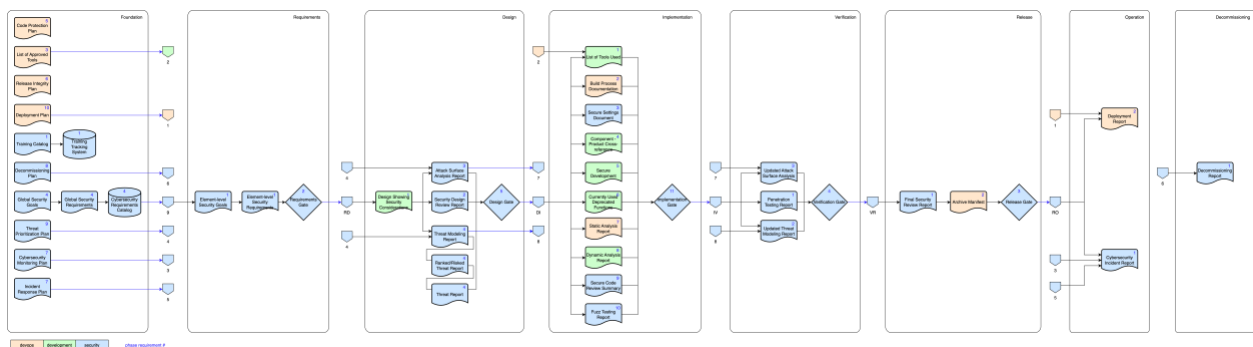


Figure 1 - AVCDL phase dependencies

For more information regarding the interpretation of these diagrams, refer to the [Understanding the Phase Product Dependencies Graph](#) secondary document.

The following pages provide a more document friendly view for each of the AVCDL phases.

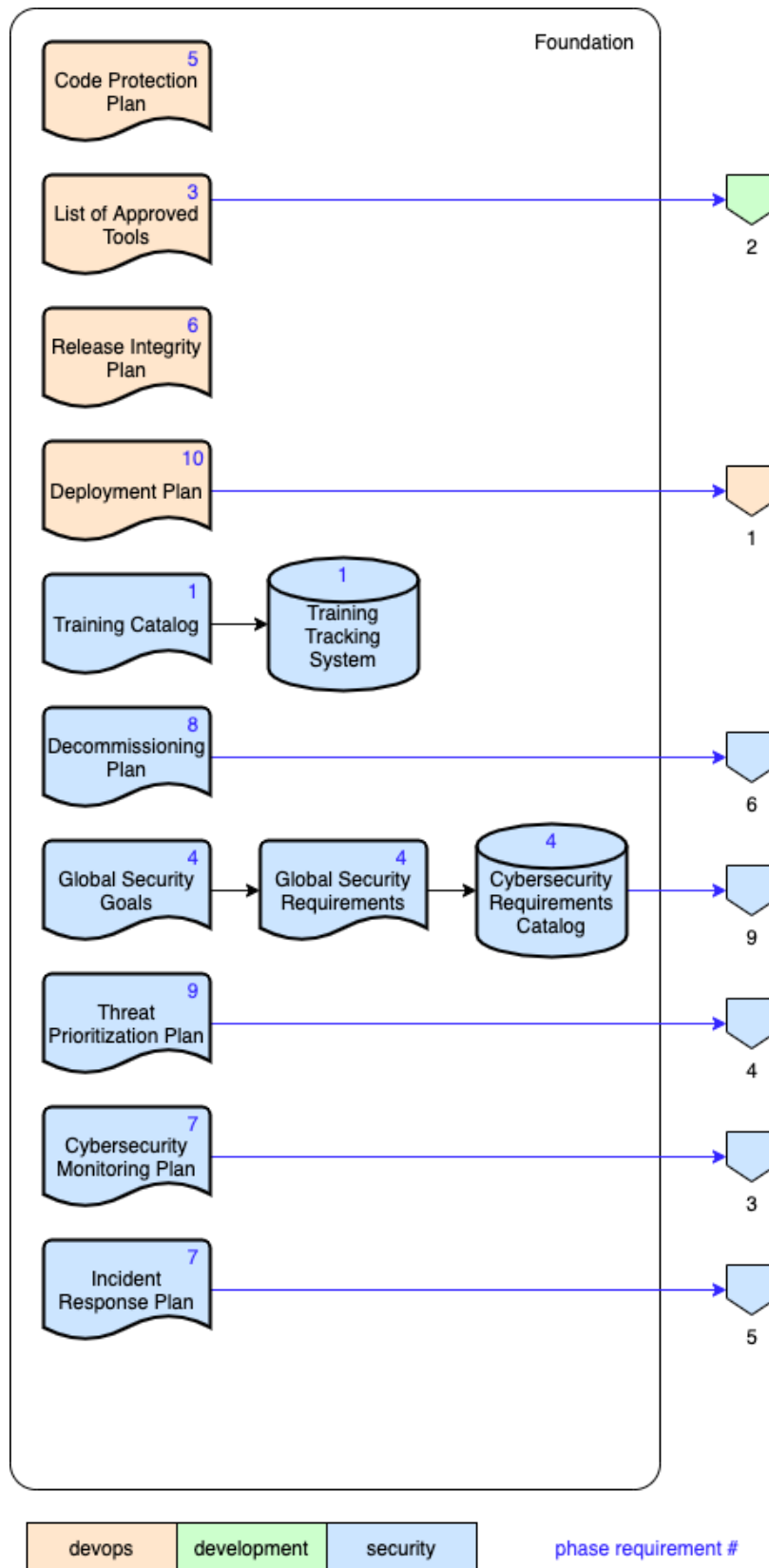


Figure 2 - AVCDL product dependencies – foundation phase

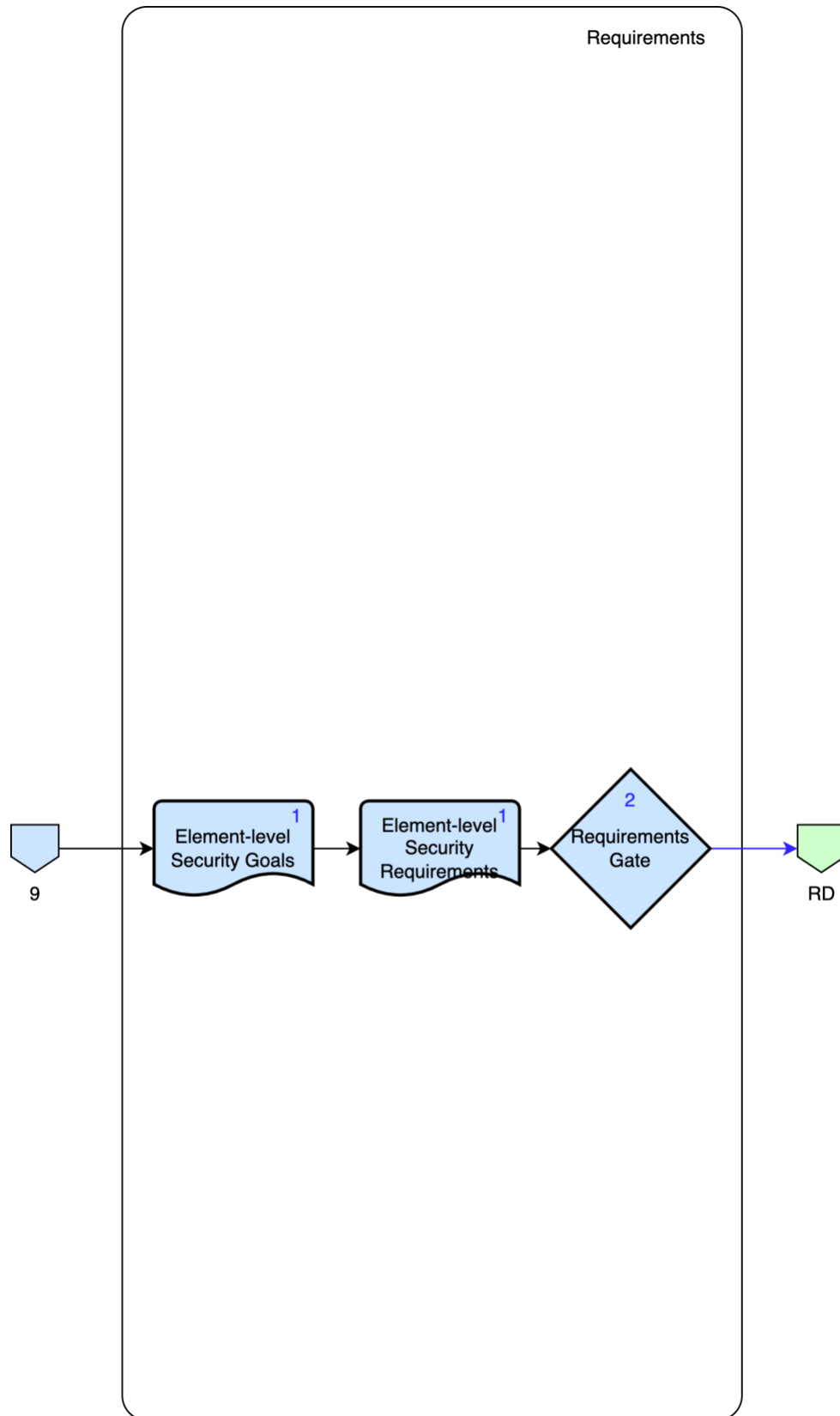


Figure 3 - AVCDL product dependencies – requirements phase

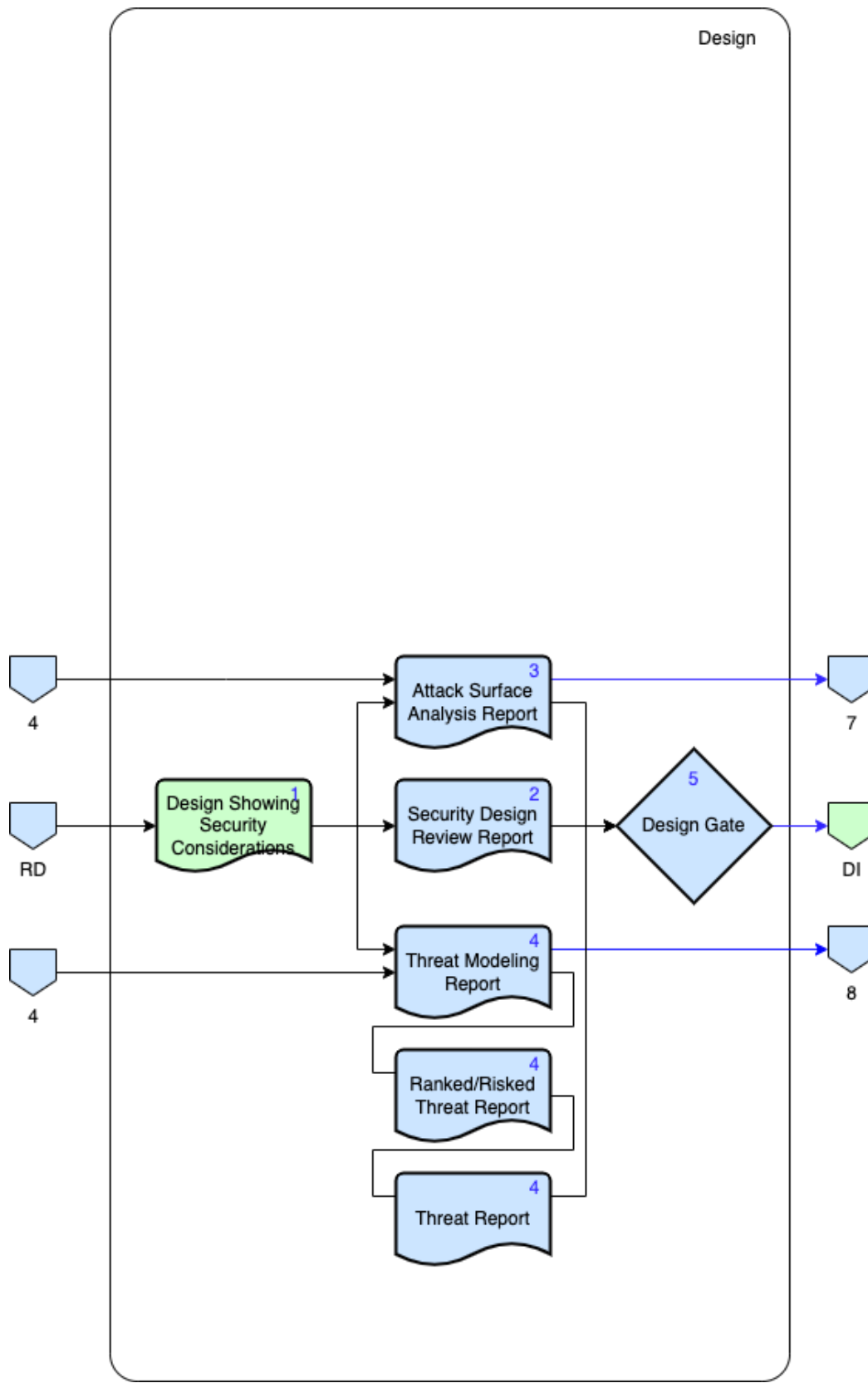


Figure 4 - AVCDL product dependencies – design phase

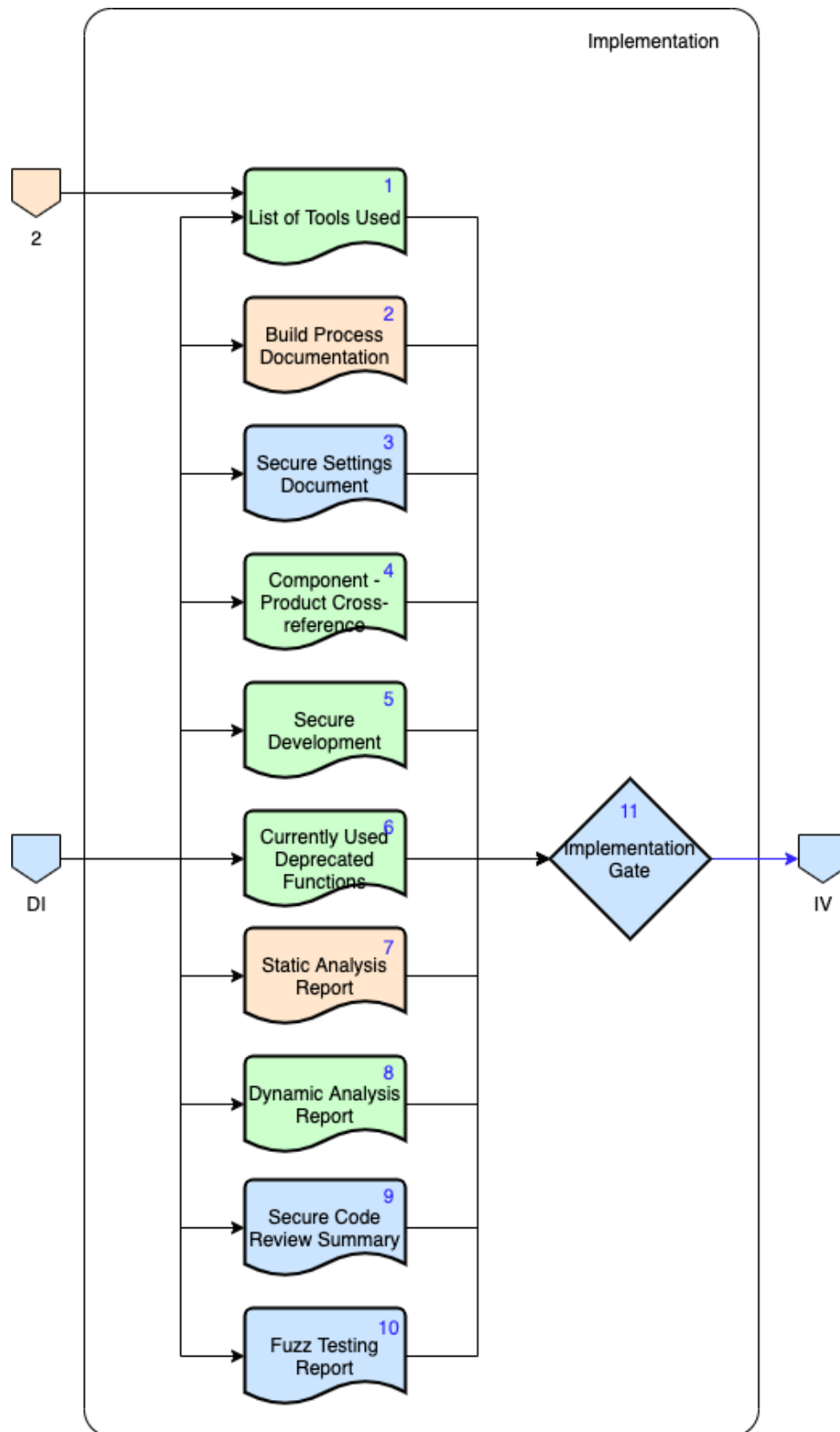


Figure 5 - AVCDL product dependencies – implementation phase

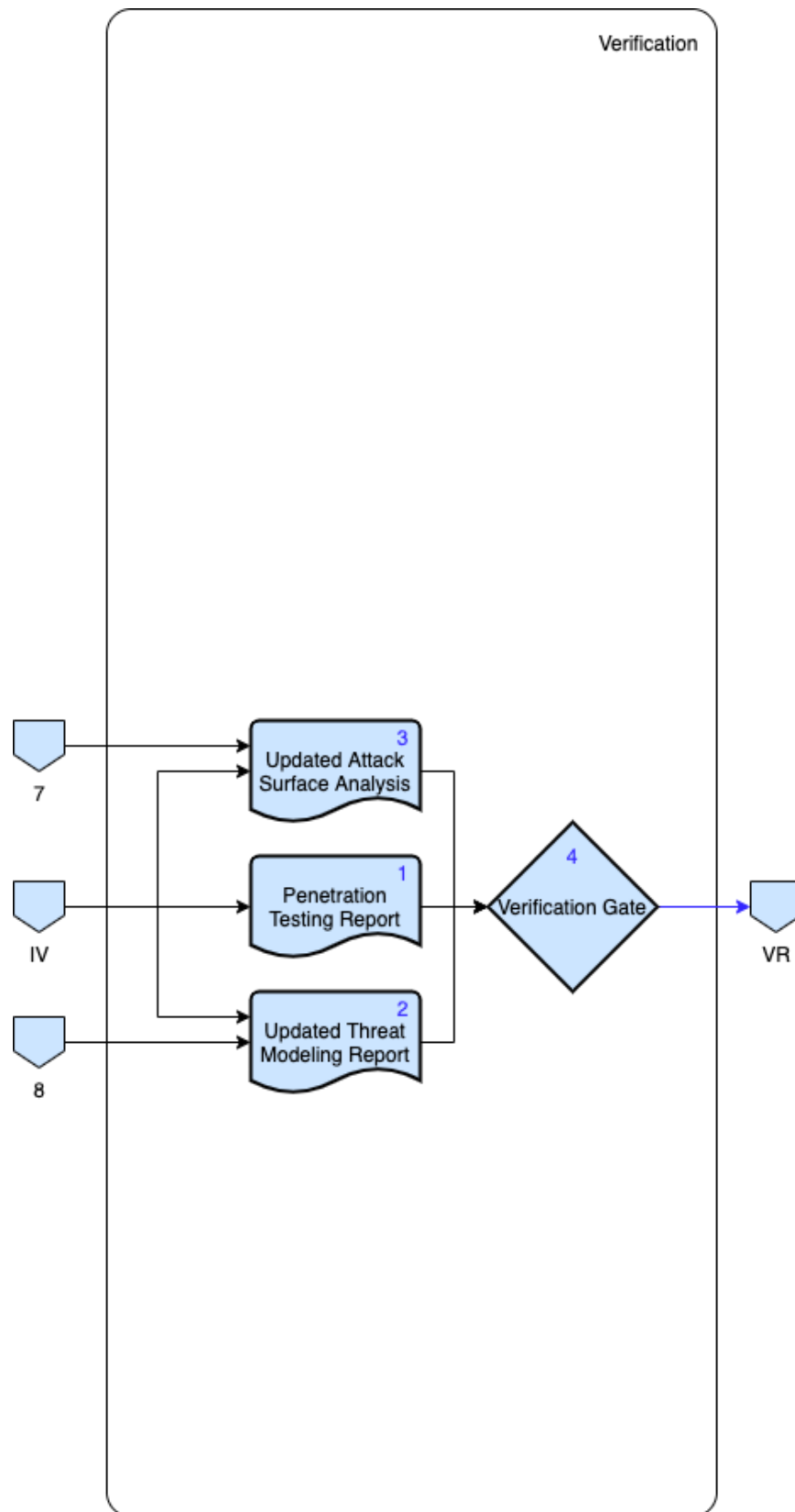


Figure 6 - AVCDL product dependencies – verification phase

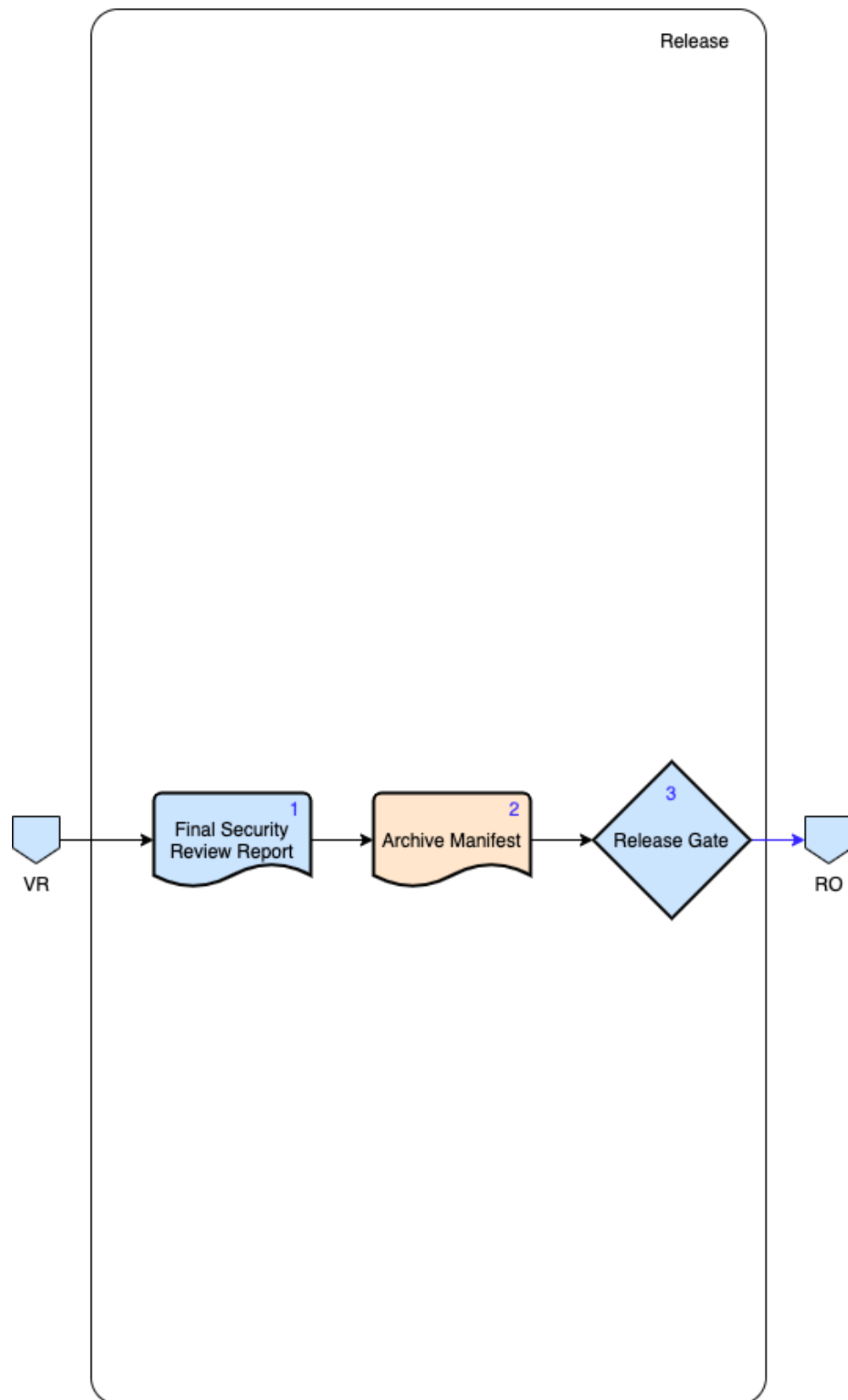


Figure 7 - AVCDL product dependencies – release phase

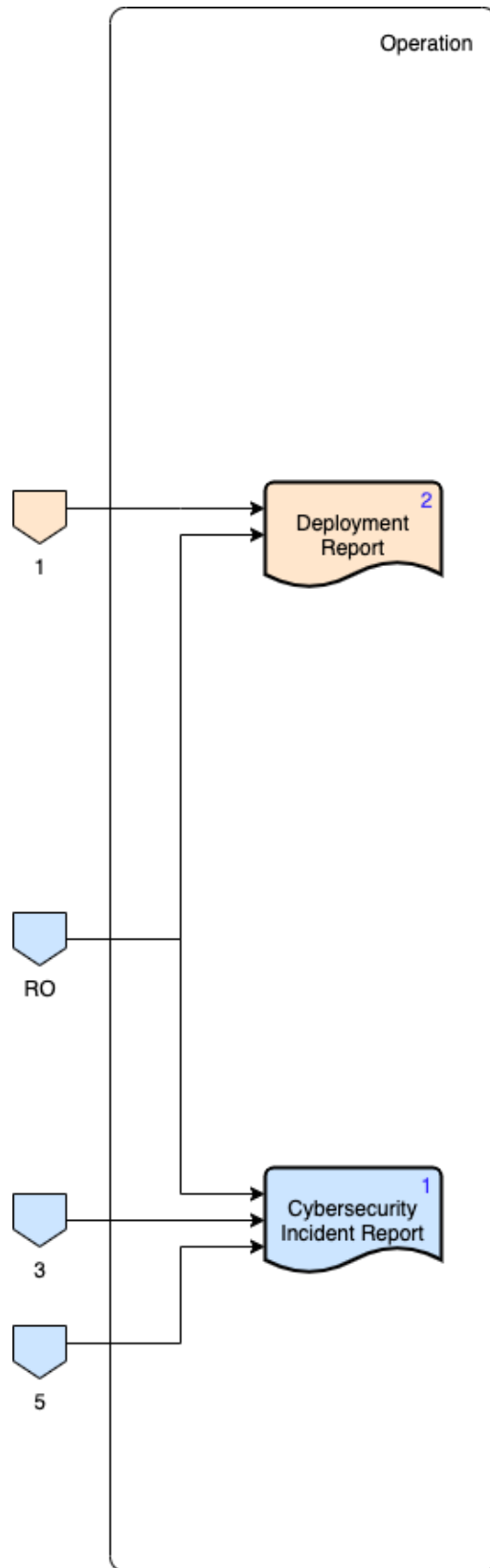


Figure 8 - AVCDL product dependencies – operation phase

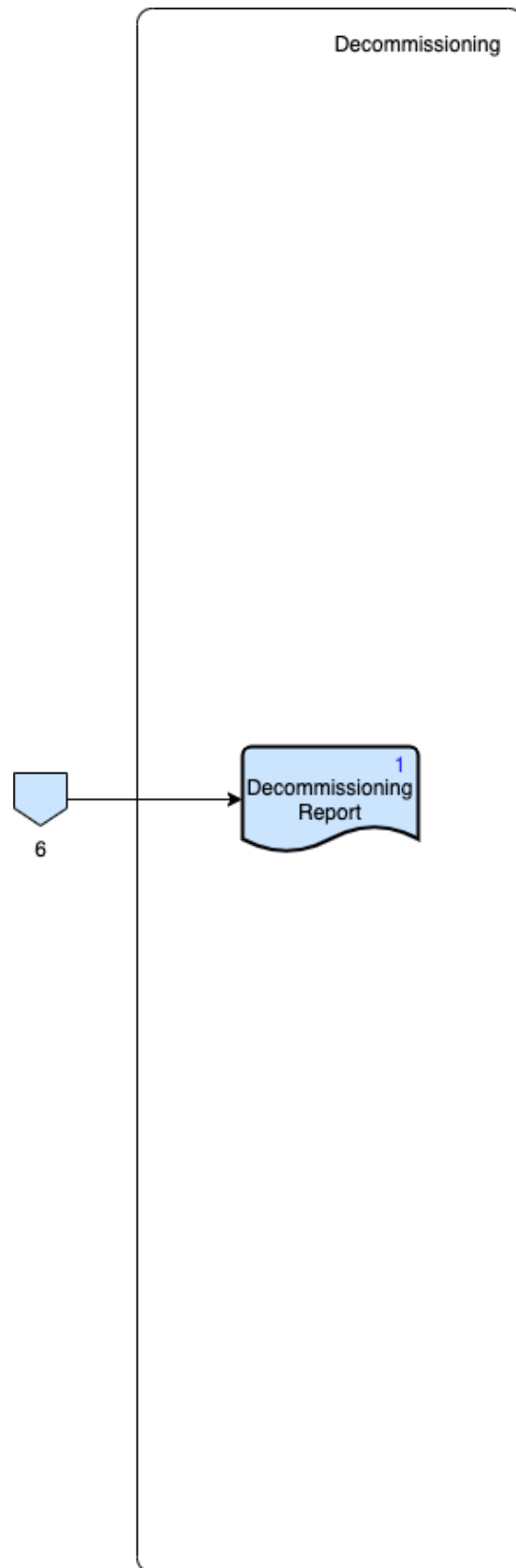


Figure 9 - AVCDL product dependencies – decommissioning phase